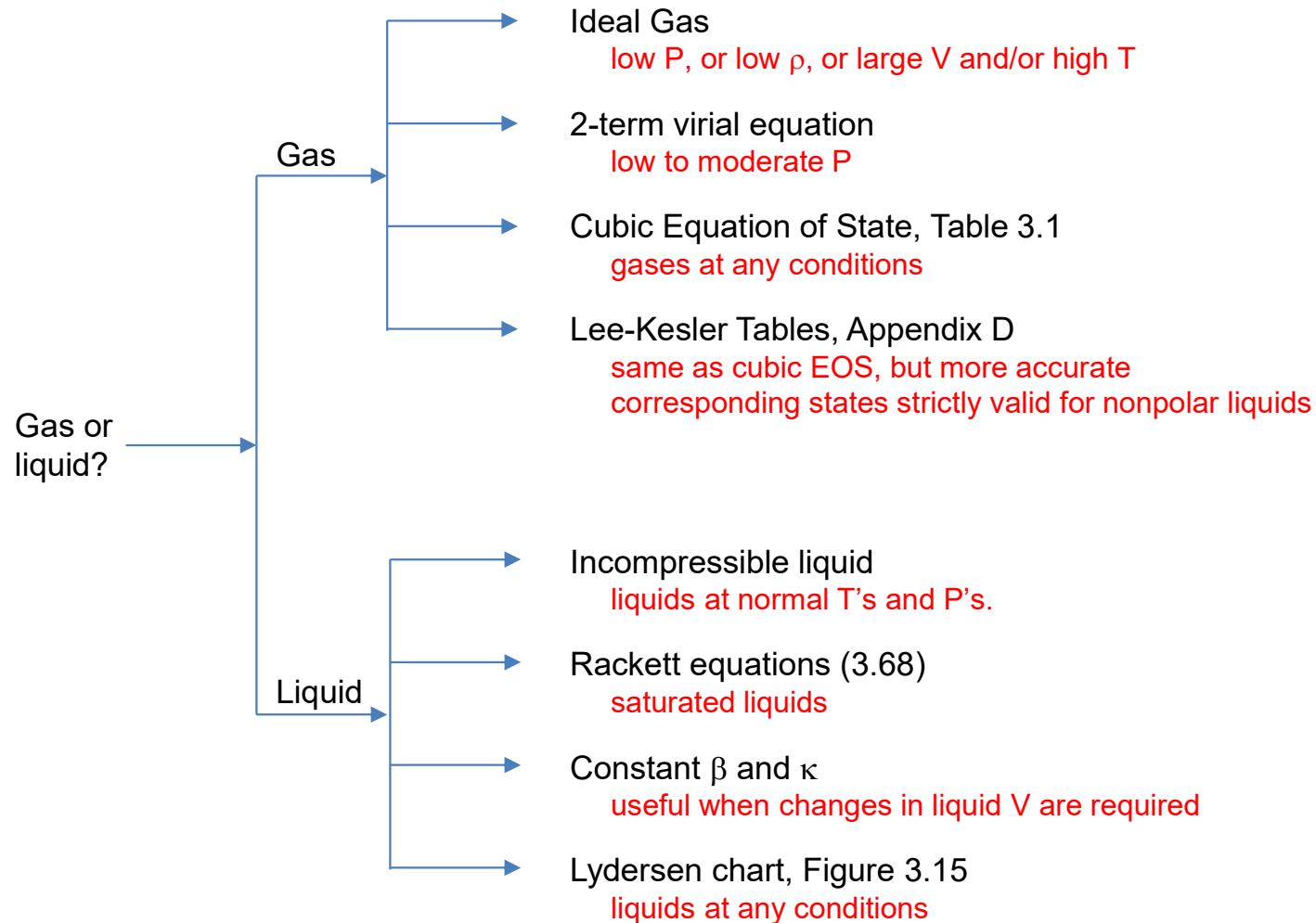


CH365 Chemical Engineering Thermodynamics

Lesson 15 Review

Equation of State Decision Tree



Homework

2nd Virial Coefficients

Slide 4

Can be Derived from Theory

2-molecule pair-wise IMFs



Electrostatic forces
(H atom + H atom $\rightarrow$ H₂)
London dispersion forces
(He, Ne, Ar, Kr, Xe)

Problem 3.78

The *Boyle temperature* is the temperature for which:

$$\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_T \equiv 0$$

- (a) Show that the second virial coefficient B is zero at the Boyle temperature.
- (b) Use the generalized correlation for B , Eqs. 3.58 to 3.62, to estimate the reduced Boyle temperature for simple fluids.

Auxiliary Forms of Virial Equations

Infinite Series – Truncate to 2 or 3 terms

L12 slide 12

$$Z = 1 + B'P + C'P^2 + D'P^3 + \dots \quad (3.33)$$

$$Z = 1 + \frac{B}{V} + \frac{C}{V^2} + \frac{D}{V^3} + \dots \quad (3.34)$$

Gases and vapors at low to moderate pressures

$$Z = 1 + \frac{BP}{RT} = 1 + \frac{B}{V} \quad (3.36, 3.37)$$

2-term truncation preferred form (linear zone on slide 10)

Gases and vapors below critical point

$$Z = 1 + \frac{B}{V} + \frac{C}{V^2} \quad (3.38)$$

3-term truncation preferred form (higher-order forms on slide 10)

(3.12 is far superior to 3.11)
(for iterative calculations)

$B = \textcircled{B}$

$$B' = \frac{B}{RT}$$

(3.35a)

$$C' = \frac{C - B^2}{(RT)^2}$$

(3.35b)

$$D' = \frac{D - 3BC + 2B^3}{(RT)^3}$$

(3.35c)

Pitzer Correlations

L13 slide 5

For next lesson; needed for Problem 3.44(b)

2nd Virial Coefficient

$$Z = 1 + \frac{BP}{RT} = 1 + \hat{B} \frac{P_r}{T_r} \quad (\text{Eq. 3.57 and 3.36})$$

$$\text{where } \hat{B} = \frac{BP_c}{RT_c} \quad (\text{Eq. 3.58})$$

Pitzer and coworkers

$$\hat{B} = B^0 + \omega B^1$$

(Eq. 3.59)

What is ω for simple fluids?

$$B^0 = 0.083 - \frac{0.422}{T_r^{1.6}} \quad (\text{Eq. 3.61})$$

$$B^1 = 0.139 - \frac{0.172}{T_r^{4.2}} \quad (\text{Eq. 3.62})$$

$$B = 0$$

$$\therefore \hat{B} = 0$$

3rd Virial Coefficient

$$Z = 1 + B\rho + C\rho^2 = 1 + \hat{B} \frac{P_r}{T_r Z} + \hat{C} \left(\frac{P_r}{T_r Z} \right)^2 \quad (\text{Eq. 3.63 and 3.38})$$

$$\text{where } \hat{C} \equiv \frac{CP_c^2}{R^2 T_c^2}$$

(Eq. 3.64)

$$\hat{C} = C^0 + \omega C^1$$

(Eq. 3.65)

$$C^0 = 0.01407 + \frac{0.02432}{T_r} - \frac{0.00313}{T_r^{10.5}} \quad (\text{Eq. 3.66})$$

$$C^1 = -0.02676 + \frac{0.05539}{T_r^{2.7}} - \frac{0.00242}{T_r^{10.5}} \quad (\text{Eq. 3.67})$$